Correction: the model DOES represent DD's frontier

It just never develops. Previous deck (2026-10-21) said the signal was absent.

What changed

The frontier probe was scored with balanced accuracy, which thresholds a BCE-trained probe's logits at 0. On a 15%-positive target that threshold is badly placed, so the metric reported a floor. Rescored on AUROC:

frontier_mask peak:	0.679	-> 0.863	(seed std <= 0.0014)
best label-only baseline, with the graph:			0.783
layer-0 control:	0.571	-> 0.676	(lift +0.186)

The model carries more frontier information than DD's own labels plus node degree can supply. The earlier 'weak evidence' reading was an artifact.

What survives

0.863 is reached by EPOCH 5, when solve_rate is 0.007, and is 0.864 at epoch 900, when solve_rate is ~0.74. Present before competence, flat through it. next_color is still the only target that tracks solve_rate (0.327 -> 0.675).

What is new

Training RELOCATES the computation instead of strengthening it: peak layer 8 -> 5, early layers rise (L1 0.708 -> 0.776), peak value unchanged.
And depth does cost solve_rate at fixed density (page 5) - against my own pre-registered prediction of 'flat'.

Same probes, same splits, two metrics

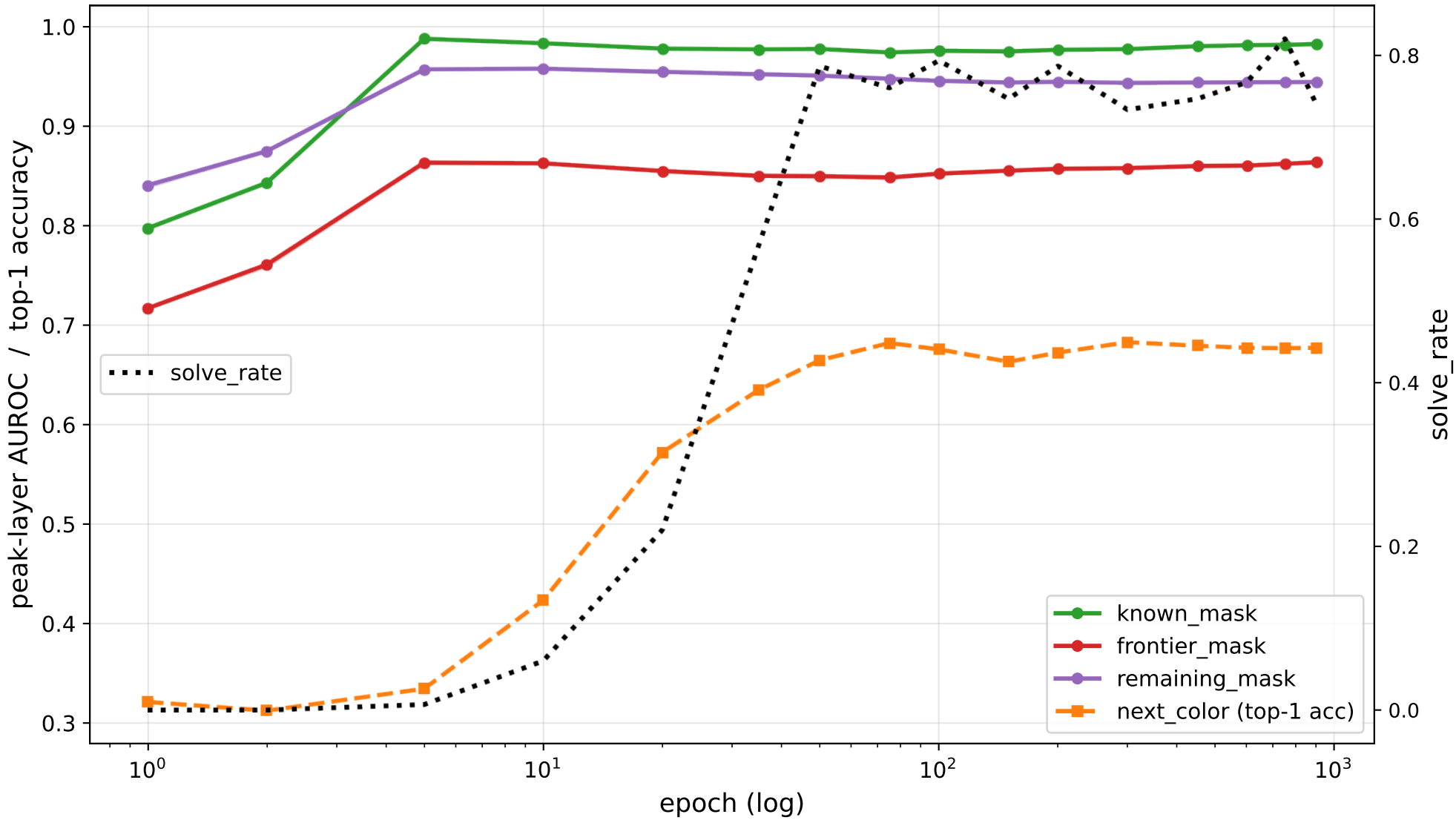
Only the scoring changed. known_probe moving 0.588 -> 0.767 is the demonstration.

predictor of frontier_mask	balanced acc	AUROC	what it knows
MODEL, best layer	**0.679**	**0.864**	**residual stream**
not_known (1 - known_mask	0.734	0.734	bookkeeping, as a 0/1 rule
known_probe (known bits onl	0.588	0.767	bookkeeping, fit as a probe
degree_aug (known + n_known_	0.648	0.783	bookkeeping + graph degree
shuffled (chance control)	0.500	0.510	nothing (permuted)

A 0/1 rule's AUROC equals its balanced accuracy by construction (not_known, row 2),
which is what keeps the hand baselines comparable across the metric change.
Average precision agrees: model 0.47 vs degree_aug 0.46 vs base rate 0.149.

Present before competence, flat through it

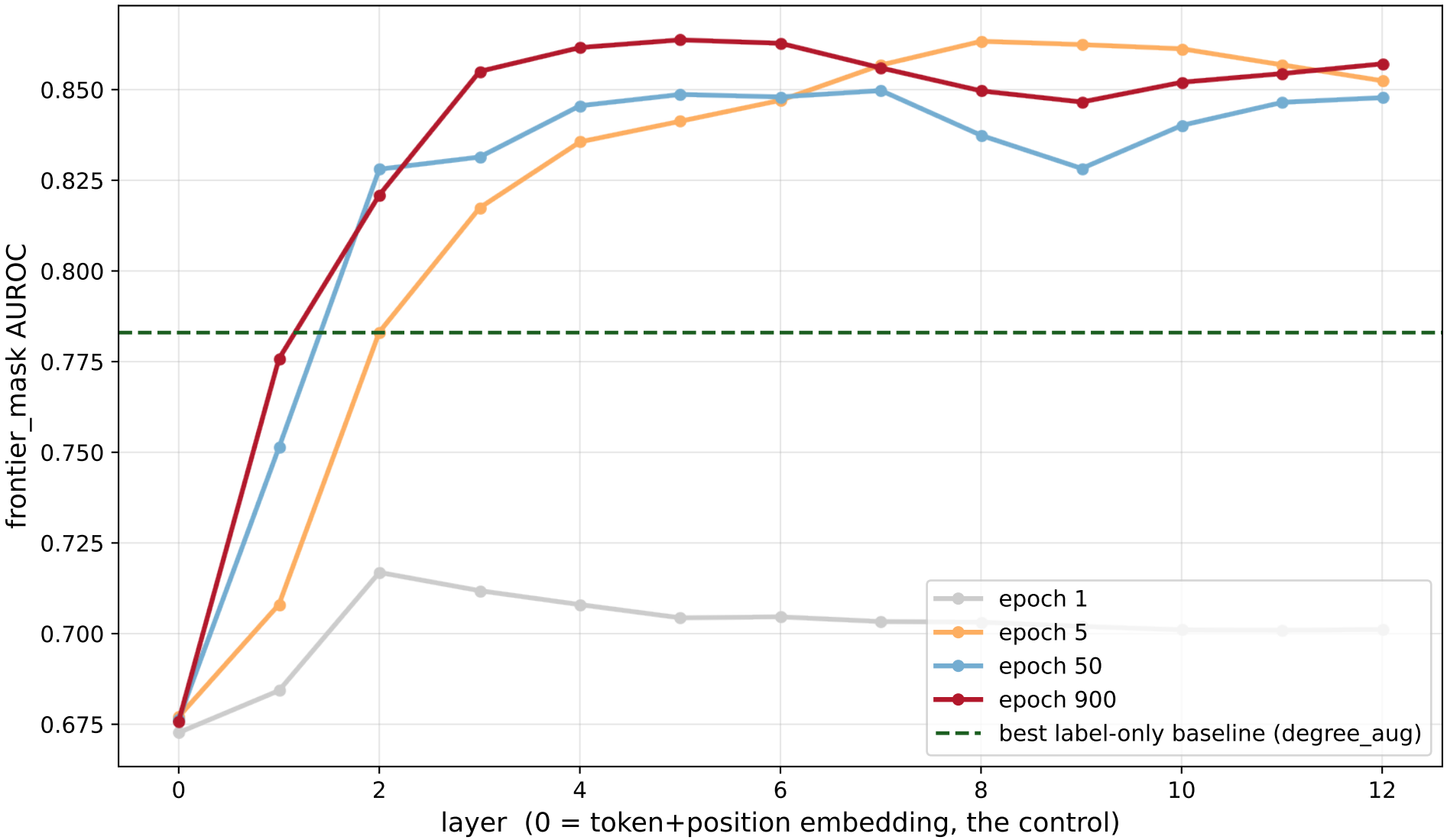
Peak-layer AUROC per snapshot, mean of 3 seeds. Shaded band = seed min/max.



frontier_mask is at 0.863 by epoch 5 (solve_rate 0.007) and 0.864 at epoch 900.
The seed band is thinner than the line: this flatness is not a noise floor.

Training moves the computation earlier, not higher

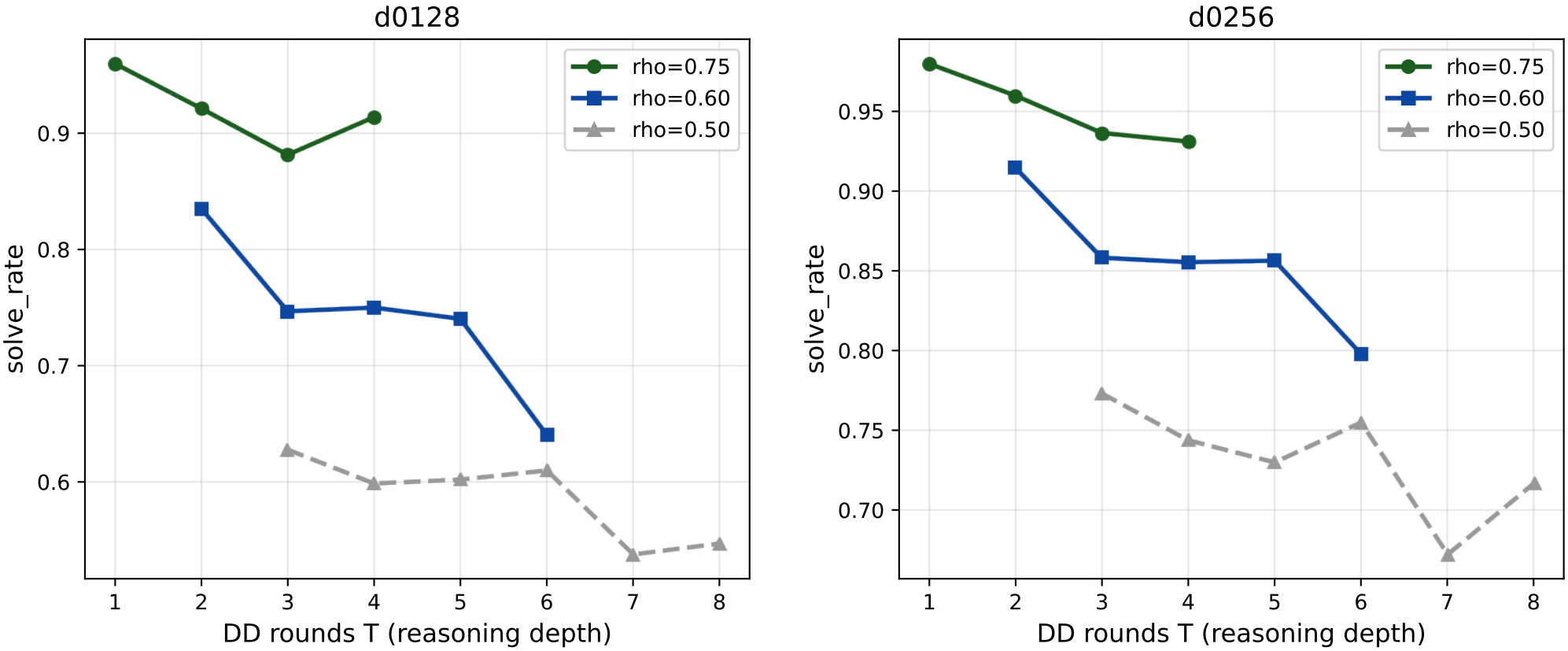
frontier_mask AUROC by layer. The peak value barely changes; its position does.



Layer 0 sits at 0.676 at every epoch, so the +0.186 lift is computed, not read off the tokenization. Peak layer 8 (ep 5) -> 5 (ep 900); L1 0.708 -> 0.776.

Depth costs solve_rate even at fixed hint density

Against my pre-registered prediction of flat. rho pinned; bins differ only in T.



Why this is not the old rho sweep in disguise:

```
rho=0.75  density IDENTICAL in every bin (spread 0.0000)   r(solve,T) = -0.71 d0128 / -0.98 d0256
rho=0.60  realized spread 0.0012 over 24 nodes             r(solve,T) = -0.91 d0128 / -0.90 d0256
rho=0.50  spread 0.0123 AND r(realized_rho,T) = -0.82   <- DISCOUNT: the confound moved inside
```

Six of six (rho, checkpoint) cells negative. Single bin gaps are only ~1.5-2.5 sigma at n = 58-698, so the consistency across cells is the evidence, not any one comparison.

Where this leaves the question

Two results that do not sit comfortably together, which is the useful part.

The tension

The frontier is represented from epoch 5 and never changes (page 3), yet solve_rate falls with reasoning depth at pinned density (page 5). A model with a static, already-good frontier representation should not care how many DD rounds an instance needs - unless the cost is somewhere else.

Candidate, and how to test it

The cost is in CHAINING COLORS, not in locating the frontier. next_color is the one target that develops with competence (0.327 -> 0.675), so a depth cost should show up as next_color accuracy falling with position-within-T, while frontier_mask AUROC stays flat across the same positions.

Training-side status, same eval throughout

best agg_solve_rate	0.847	d0256 @300ep (unchanged, still leads)
epochs 300 -> 900	0.817	predicted not to help; it hurt but raised solve_rate_k0 0.24 -> 0.87
n_train 5000->20000	running	
d_model 256 -> 384	running	

Caveats

One training trajectory; the 3 seeds vary the PROBE, not the training run. Depth bins come from one instance pool per density, one seed each.